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Project Title:	Gender Related Differences In Central Autonomic Control Following Simulated Microgravity In Rats		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
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Space Biology Special Category:	None		
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Task Description:	Prolonged bedrest or exposure to reduced gravity, as occurs during space flight, results in cardiovascular deconditioning. In astronauts, upon returning to the normal gravity of earth, there is a decreased ability to exercise and a greater tendency to faint when standing. Normally, when a person stands from a sitting position, heart rate increases and increased nerve activity (sympathetic nerve activity) to blood vessels causes vasoconstriction, so that blood is returned to the heart and doesn't pool in the legs. Following cardiovascular deconditioning the normal compensatory vasoconstriction is not as pronounced, and blood isn't returned to the heart as effectively. This can result in light-headedness and fainting. Interestingly, following space flight this tendency to faint upon standing is more pronounced in female, compared to male, astronauts. Previously, using an animal model for cardiovascular deconditioning, the hindlimb unloaded rat, we demonstrated that in both males and females the ability to increase sympathetic nerve activity when blood pressure drops is attenuated following deconditioning (14 days of hindlimb unloading). When age-matched male and female control animals were evaluated, we found that, compared to males, female rats exhibited less of a compensatory increase in vasoconstrictor sympathetic nerve activity when blood pressure was lowered. Thus, the combined effect of sex and cardiovascular deconditioning resulted in less ability to defend against a drop in blood pressure in the females. Since baseline heart rate is higher in females compared to males, and since cardiovascular deconditioning results in an increase in heart rate, we considered that heart rate might be different between males and females and might be affected differently by cardiovascular deconditioning. Heart rate is determined by the balance between sympathetic nerve activity to the heart, which would increase heart rate, and parasympathetic nerve activity, which slows the heart. In control rats, females had a higher resting heart rate compared to males and the resting heart rate in females was more dependent on sympathetic nerve activity to the heart. Baseline heart rate increased in both male and female rats following cardiovascular deconditioning and it was due to both decreased parasympathetic and increased sympathetic influences on the heart in both sexes. However, cardiovascular deconditioning did not affect the compensatory increase in heart rate when blood pressure was decreased in either males or females. Therefore, it appears that in response to cardiovascular deconditioning, a decreased ability to maintain blood pressure in females is predominantly due to attenuated increases in sympathetic nerve activity to the blood vessels. When blood pressure drops, during an orthostatic challenge (standing) or due to blood loss (hemorrhage), humoral systems also contribute to the compensatory response. Vasopressin, a hormone which is released when blood pressure drops, has direct effects on blood vessels to cause constriction and therefore might work to counteract the initial decrease in blood pressure. We found that baseline levels of vasopressin are elevated in hindlimb unloaded rats, and this effect is greater in males compared to females. In addition, when rats are exposed to progressive hemorrhage, the progressive decrease in blood pressure is accompanied by a progressive elevation in plasma vasopressin. The response to blood loss is complex and future experiments are needed to determine the role of heart rate and humoral changes. Orthostatic intolerance, or the inability to maintain adequate cardiac output and cerebral blood flow while standing, is a common feature of cardiovascular deconditioning following spaceflight or prolonged bedrest. Interestingly, the incidence of presyncope (lightheadedness, loss of peripheral vision, sudden drop in systolic pressure) following space flight is much greater in female compared to male astronauts. Alterations in blood volume, cardiac function, vascular function, and neurohumoral control mechanisms have all been implicated in orthostatic intolerance following cardiovascular deconditioning. Normal adjustments to changes in posture require appropriate neurohumoral control mechanisms, which likely differ in females compared to males. Hindlimb unloading (HU) in rodents is an animal model of cardiovascular deconditioning, and produces cardiovascular effects similar to those observed in humans. No studies have critically evaluated potential differences in the central nervous mechanisms controlling autonomic function that may contribute to sex differences in cardiovascular deconditioning. The general hypothesis to be tested is that: Compared to male rats, female rats will exhibit greater signs of orthostatic intolerance following hindlimb unloading. These effects are centrally mediated, involve the neurotransmitter GABA in central autonomic pathways, and differences are associated with ovarian hormone levels in the females. Information gained from these experiments will be applicable to the design of ground based studies in humans. The ultimate goal is to understand the mechanisms for male/female differences and to design specific countermeasures to improve cardiovascular function upon return to earth's gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ground-based animal model used in these experiments produces cardiovascular deconditioning similar to that seen in astronauts following space flight. Similar deconditioning occurs following prolonged bedrest in humans. In fact, bedrest is a ground-based model used to study cardiovascular deconditioning in humans. With the aging population in the Unitied States, and the likely increase in periods of prolonged bed-rest, any information gained from these experiments will be applicable to possible measures to limit cardiovascular deconditioning following bed-rest.
Task Progress:	During the last year of funding we have completed experiments on the effects of progressive blood loss in male and female, control and hindlimb unloaded rats (animal model for cardiovascular deconditioning as occurs following space flight). Hindlimb unloading resuted in elevated baseline vasopressin levels, especially in males. However, in general, the hemorrhage associated increase in vasopressin was delayed in hindlimb unloaded compared to control rats. Contrary to our expectation, neither female sex nor hindlimb unloading appeared to compromise maintenance of mean arterial blood pressure during progressive hemorrhage in conscious rats. An unexpected finding was that hemorrhage was associated with a pronounced arrythmia (characterized by periodic profound decreases in heart rate) that was much more prevalent in male compared to female rats, and was greatly decreased by hindlimb unloading. This raises the possibility that inhibitory motor drive (parasympathetic nerve activity) to the heart is affected by hindlimb unloading. Future experiments are planned to further evaluate this interesting observation.
Bibliography Type:	Description: (Last Updated: 01/15/2021)
Abstracts for Journals and Proceedings	Foley CM, Weissbach SP, Hernandez MJ, Price EM, Heesch CM. "GABAA receptor alpha1, alpha2, gamma1, gamma2, gama3, and delta subunit mRNA expression in hindbrain nuclei in nonpregnant and pregnant rats." Experimental Biology 2005. FASEB Journal 2005 Apr;19: A615. , Apr-2005

Abstracts for Journals and Proceedings	Heesch CM, Haught AL, Phaup JG, Foley CM, Mueller PJ, Rossi NF, Hasser EM. "Effects of cardiovascular deconditioning on plasma vasopressin (AVP) levels during blood loss in male and female rats." Experimental Biology 2005. FASEB Journal 2005 Apr;19: A614. , Apr-2005
Articles in Other Journals or Periodicals	Foley CM, Mueller PJ, Hasser EM, Heesch CM. "Hindlimb unloading and female gender attenuate baroreflex-mediated sympathoexcitation." In Press , Nov-2005